

PROGRESS STATION, Antarctica

WMO: 895740

Lat: 69.378S

Lon: 76.391E

Elev: 48

StdP: 14.67

Time Zone: 5.00 (E05)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-28.5	-22.1	-35.0	0.8	-24.6	-31.1	1.0	-18.6	46.2	12.0	39.7	16.1	1.9	300	1.089

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.2	40.5	33.4	38.8	32.4	37.4	31.5	34.1	39.2	33.0	37.7	32.0	36.4	6.1	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.6	24.8	33.2	28.8	22.9	32.8	27.2	21.2	32.8	12.7	39.4	12.3	38.0	11.9	36.5	38.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
36.1	29.6	24.9	-32.5	42.6	9.9	2.4	-39.7	44.3	-45.5	45.7	-51.0	47.1	-58.2	48.8		
			-32.0	35.6	8.7	1.2	-38.2	36.5	-43.3	37.2	-48.2	37.9	-54.6	38.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.3	32.7	26.6	16.4	9.3	5.8	4.7	2.7	3.6	4.8	10.7	24.0	31.6
	DBStd	13.88	2.86	4.32	7.29	8.18	10.59	10.11	12.74	11.27	10.77	7.86	5.90	3.72
	HDD50	13015	535	654	1043	1222	1371	1359	1466	1438	1355	1219	781	572
	HDD65	18490	1000	1074	1508	1672	1836	1809	1931	1903	1805	1684	1231	1037
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	9.8	7.0	9.7	10.8	9.9	11.0	10.9	10.1	11.1	10.5	9.8	9.2	7.6
Precipitation	PrecAvg	13.2	1.5	1.5	1.3	0.8	0.9	0.9	1.1	0.9	0.8	0.9	1.3	1.2
	PrecMax	17.7	4.2	2.8	2.7	2.1	2.1	2.2	2.9	3.1	2.1	2.0	3.3	2.4
	PrecMin	6.7	0.5	0.7	0.4	0.3	0.3	0.2	0.0	0.1	0.0	0.2	0.5	0.4
	PrecStd	2.0	0.7	0.6	0.6	0.4	0.4	0.4	0.6	0.6	0.4	0.4	0.7	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.8	39.5	32.9	29.1	31.6	28.2	29.1	28.2	27.5	29.7	36.6	42.7
		MCWB	35.9	32.5	28.7	25.4	27.0	24.8	25.2	25.9	24.0	24.5	30.3	35.1
	2%	DB	40.7	35.8	29.6	25.1	26.1	24.2	24.9	23.4	24.2	26.4	35.0	40.7
		MCWB	33.6	29.7	25.8	22.6	23.4	21.5	22.0	20.4	21.4	22.4	29.5	33.6
	5%	DB	39.0	34.2	27.7	23.0	23.4	20.8	22.2	20.5	21.7	24.3	33.4	39.1
		MCWB	32.6	29.1	24.4	20.7	20.8	18.3	19.9	17.8	18.8	21.0	28.3	32.4
	10%	DB	37.7	32.7	25.9	20.6	20.4	18.0	19.1	17.9	19.2	22.1	31.9	37.6
		MCWB	31.9	28.3	23.1	17.9	18.1	15.7	17.0	15.7	16.4	19.1	27.5	31.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.3	33.4	29.7	26.2	27.1	25.1	25.5	25.9	24.6	25.3	31.9	35.8
		MCDB	42.4	36.6	32.8	28.4	30.1	28.0	27.8	27.4	26.6	27.9	35.0	41.7
	2%	WB	34.3	31.5	26.7	23.2	23.7	21.9	22.8	20.7	21.6	23.0	30.3	34.1
		MCDB	39.2	34.0	28.5	25.2	26.0	24.0	24.5	22.9	24.2	25.8	33.5	39.7
	5%	WB	33.3	30.1	24.9	20.8	21.0	18.6	19.9	18.1	19.2	21.1	29.2	32.9
		MCDB	37.9	33.3	27.1	22.9	23.2	20.6	22.0	20.2	21.4	24.1	32.7	38.1
	10%	WB	32.4	28.6	23.3	18.0	18.3	15.9	17.1	15.9	16.5	19.2	27.8	31.9
		MCDB	36.8	32.3	26.0	20.4	20.5	18.1	19.2	18.1	19.0	22.1	31.6	36.7
Mean Daily Temperature Range	5% DB	MDBR	6.2	5.7	6.0	7.1	8.1	8.0	9.0	9.4	8.0	7.8	7.3	6.5
		MCDBR	7.1	6.7	5.5	7.2	7.2	9.4	8.0	8.5	7.9	8.2	7.5	7.7
		MCWBR	5.5	5.8	5.1	7.0	6.7	8.8	7.5	8.2	7.3	7.5	6.5	5.8
	5% WB	MCDBR	6.5	5.4	5.8	7.4	7.1	9.9	8.0	9.2	8.6	8.0	6.8	7.1
		MCWBR	5.4	5.5	6.0	7.4	7.0	9.5	7.6	9.2	8.1	7.7	6.1	5.5
Clear-Sky Solar Irradiance	taub	0.257	0.246	0.219	0.162	0.121	N/A	0.124	0.158	0.202	0.217	0.242	0.258	
	taud	2.314	2.344	2.305	2.103	1.903	N/A	1.884	2.060	2.221	2.235	2.245	2.262	
	Ebn at Noon	315	299	272	206	29	0	19	202	274	308	318	319	
	Edh at Noon	34	28	21	11	1	0	1	11	22	30	36	37	
All-Sky Solar Radiation	RadAvg	2552	1660	813	237	19	0	3	119	592	1431	2354	2881	
	RadStd	92	78	38	11	1	0	0	7	31	52	80	102	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page